Base RSR Training Data

Construction and validation of the train set (PMB 5.1.0)

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1 At a Glance

This document reports how I generated base replacement self-repair (RSR) data with an open-weights language model, starting from the gold train split of PMB 5.1.0. "Base" means that the repairs carry no editing expression such as "I mean" or "actually". These will be added by script in a later step.

The training set now holds 3,489 RSR items, about 500 in each of seven categories, against 9,486 fluent PMB train sentences, a ratio of about 1 : 2.7.[1] This experiment shows that, for this project, generating RSR data with an LLM is a workable method. Six of the seven categories meet my quality thresholds through generation and automatic checks alone, and the seventh, edge label, meets them after I selected its items by hand. I will therefore build the dev and test sets in the same way. Before that, the next steps are to add editing expressions and to compare the generated repairs with repairs in Switchboard.

Section 2 explains why I moved from rules to an LLM, and Section 3 shows what an item looks like. Section 4 and Section 5 describe how the items were generated and how I judged their quality. Section 6 gives the results. The last section covers limitations and next steps, and closes with three questions on which I would like your advice (Section 7.1).

2 Why I do this

My first data pipeline was entirely rule-based. Each RSR category needed its own transformation script, so I had five or six scripts to maintain, and some categories, such as tense, still needed items written by hand. During my research proposal presentation, Prof Koller suggested that I consider generating the data with an LLM instead. Given the workload of the rule-based system and the number of places where it could go wrong, the suggestion made a lot of sense.

My aim is therefore two-fold, and the second part depends on the first. I first want to find out how well an LLM, run on the computing resources available to me, can generate RSR data. Quality has two sides in this task, naturalness and correctness, and correctness applies to both ends, the sentence and its SBN. If the quality is good enough, I will then make LLM generation the main data source of the project.

Two design principles follow from these aims, and they run through the rest of the document. The first is about generation. I give the model as much freedom as possible, so that the results measure the model's own ability. It is constrained only by its input, a PMB sentence with its SBN and the category to produce, and by a single tool, a WordNet lookup (Section 4). The second is about quality control, which follows the rule "selection, not correction". An item that fails a check is dropped and never fixed, so every item in the data is the model's own output,

[1] PMB 5.1.0 has 9,552 train documents. I hold out 66 of them that look like a repair on the surface but are not one, such as appositions ("I'm from Wellington, the capital of New Zealand"), lists, and contrasts like "not a nurse, but a doctor". They are kept for a hard-negative test and appear in no training data.

and the failure rate becomes evidence of how well the model does the task. To judge how correct and how natural the items are, I combine three kinds of evidence: automatic structural checks, manual annotation, and a comparison with repairs in Switchboard, which is still in progress.

3 What the data look like

In early versions of my research proposal, I used "replacement" and "repair" loosely for the phenomenon as a whole. From now on, RSR stands for replacement self-repair only, and "repair" means only the part of the utterance that reflects the speaker's final intention. The repair site is the place in the gold SBN that the repair changes, such as a concept, a name or the tense of a verb.

Each item is built from one PMB sentence and its gold SBN. The gold sentence supplies the repair, and the model invents the reparandum in front of it. Take the gold sentence "Hebrew is my mother tongue", which becomes "Hebrew was, is my mother tongue". The two SBNs are shown below.

```
% gold: Hebrew is my mother tongue.
hebrew.n.01
be.v.02              Theme -1 Time +1 Co-Theme +3
time.n.08            EQU now
person.n.01          EQU speaker
mother_tongue.n.01   Creator -1

% RSR: Hebrew was, is my mother tongue.
hebrew.n.01
                     CORRECTION <1
be.v.02              Theme -1 Time +1
time.n.08            TPR now
                     CONJUNCTION <2
be.v.02              Theme -3 Time +1 Co-Theme +3
time.n.08            EQU now
person.n.01          EQU speaker
mother_tongue.n.01   Creator -1
```

If the `CORRECTION` box is deleted, what remains is exactly the gold SBN of "Hebrew is my mother tongue". The notation is designed so that this holds for every item, once any placeholder concept is merged back. Because every item starts from a PMB sentence with a gold SBN, a script can test this property automatically on each item, and most of the checks in Section 5 rest on it.

A good item therefore meets three conditions. The sentence and the SBN are both correct and agree with each other; deleting the withdrawn version gives back the gold meaning; and the repair sounds natural to a fluent speaker.

3.1 The seven categories

I defined seven RSR categories, derived from my formalism. Each category is defined by what the repair replaces in the SBN, which is why the names mix linguistic terms such as tense with SBN terms such as edge label (Table 1).

Table 1: The seven RSR categories and their size in the training set. Reparanda are shown in italics.

category	what is replaced in the SBN	example	items
content word	a concept, by another of the same part of speech that is not a synonym	A baby is sleeping in *the basket*, the cradle.	500
name	the value of a name	Were these pictures taken *in France*, in Germany?	500
number	a quantity	The bank lent her *300*, 500 dollars.	500
temporal	a clock time, day, month or year	I was born *in 1971*, in 1972.	500
tense	the tense of a verb	Hebrew *was*, is my mother tongue.	500
forwarding	the subject, with the verb said again	*John hates*, I hate spiders.	500
edge label	the role between two concepts, or a place relation	He jumped *from*, into the train. / Tom was born *near*, in Boston.	489
all			**3,489**

Two categories need a word of explanation. In forwarding, the speaker restarts the clause with a new subject and says the verb again. In edge label, the repaired word is a preposition, and in SBN the change falls on the role that links two concepts, such as `Source` (“from”) and `Destination` (“into”).

With the balanced categories, the differences between them in the parsing experiments cannot come from differences in training data size.

4 Data generation setting

Figure 1 shows the whole process. This section covers generation, and Section 5 covers the checks and annotation that follow.

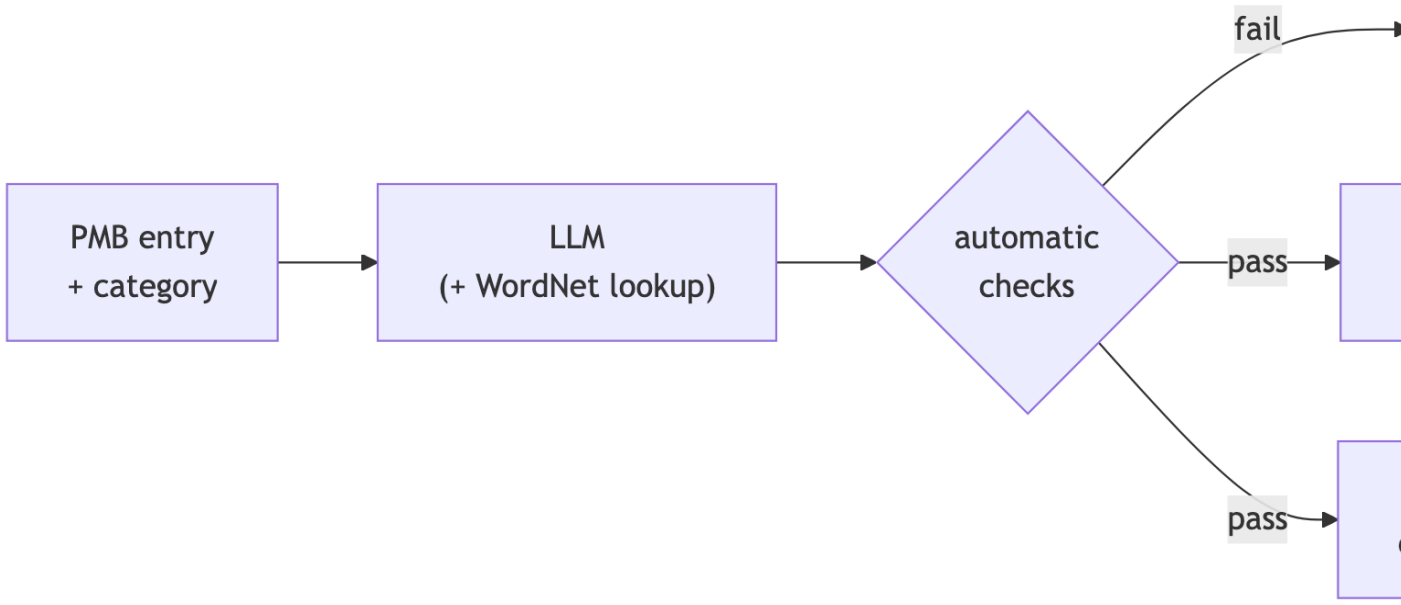


Figure 1: From a PMB entry to the training set.

From the items that pass, I drew a simple random sample of 500 per category with a fixed seed. Edge label, with fewer items kept, contributes all of its 489. Before sampling, small surface slips such as letter case and quote marks were fixed.

4.1 Model and data

I used Muse-Glimmer-30B, an open-weights model, on PMB 5.1.0 train gold. I chose it for what the task needs. It can call an external tool over several turns, which the WordNet lookup requires, and it returns its answer in a fixed structure that holds the sentence and the SBN together. As a 30B model, it can also run on a single GPU of the LST cluster, and because the

weights are open, every run can be repeated on university hardware. All runs use the same sampling settings.[2] The results in Section 6 show that the model is up to the task.

4.2 Input and prompt

Each request consists of the prompt and one PMB entry, that is, a gold sentence and its gold SBN, together with the category to produce. The prompt is shared by all categories and has three parts.

1. An SBN manual, which tells the model how to annotate a sentence in SBN. It is a stand-alone document based on Bos' papers (2021, 2023) and the annotation practice in PMB gold.
2. The repair notation, which explains how to annotate an utterance with RSR in the `CORRECTION`–`CONJUNCTION` construction. It is derived from my notation document.
3. The task, which asks the model to read both sides of the PMB entry and add an RSR of the given category to both.

Every example in the prompt was checked automatically for well-formed SBN, and the PMB documents that supply these examples are excluded from the data.

Within the given category, the model decides everything else: where the repair occurs, which word it uses as the reparandum, and how the sentence and the SBN are written. If a sentence offers no suitable site, the prompt allows the model to decline. It has one attempt per item, with no retry and no self-check, because retrying until an item passes would hide how often the model fails.

Two categories have less freedom. For some temporal items and for the place repairs in edge label, the input also names the site to repair, and these make up more than half of the final items in both categories. Temporal needed this because only 297 train documents contain a usable time expression, so many of them had to yield a second item. The reason for edge label is given in Section 6.2.

4.3 Tool calling

The model's only tool is a WordNet lookup. Concepts in SBN are WordNet 3.0 synsets, so a concept outside WordNet cannot be represented, and the model needs a way to check what it writes. The lookup lists the senses of a word with their definitions and confirms whether a concept exists. It shows no synonyms or hypernyms and offers no list of candidate replacements, so the model has to come up with the reparandum itself. I kept the tool this narrow on purpose. A list of candidates would decide in advance which slips are believable, and that judgement is part of what I want to test.

[2] Reasoning effort high, temperature 1.0, top-p 0.95, top-k 64, seed 13.

5 Quality assurance setting

Each kind of evidence answers a different question.

1. The automatic checks run on every item and ask whether it is well-formed and faithful to its gold SBN. An item that fails is dropped.
2. The manual annotation reads a sample of the items that pass and asks what the checks cannot see: whether the SBN is correct and whether the repair sounds natural.
3. The Switchboard comparison will describe how the generated repairs relate to repairs in real speech.

Whether a category can enter the data is decided by four thresholds (Table 2).

> **Caution**
>
> The figures come from my experience as an annotator, as I know of no established standard for data of this kind.

I fixed them before seeing any results, so that I could not adjust them to fit.

Table 2: Thresholds for admitting a category.

evidence	measure	threshold
automatic checks	answered items that pass every check	95%
automatic checks	sentences the model declines	3%
manual annotation	items correct in meaning	90%
manual annotation	items that sound natural	80%

5.1 Structural checks

The checks fall into five groups (Table 3).

Table 3: The five groups of automatic checks.

check group	an item fails if
well-formed graph	the SBN is not well-formed or has a cycle, a pointer lands on no concept, or a concept is not in WordNet
graph returns to gold	deleting the withdrawn version, and merging back any placeholder concept, does not give back exactly the gold SBN, or something away from the repair site has changed

check group	an item fails if
sentence and graph agree	the sentence does not match the SBN, for example because the words out of the RSR span are also changed by the model, or the model chooses a reparandum that is identical to the repair (i.e., resulting in a repetition).
site within the category	the repair site is not one the category allows, such as one place inside the other (“Osaka, Japan”)
a repair, not a synonym	the reparandum and the repair belong to the same WordNet synset, as “sofa” and “couch” do; however, a more general or a more specific word is allowed

5.2 Manual annotation

From the items that pass every check, I draw 15 per category at random and read them one at a time, without seeing the category or any check result. For each item I answer two questions: does the repaired SBN mean what the repaired sentence says, and is the repair natural? By natural I mean acceptable to a fluent speaker, that is, grammatical, recognisable as a repair, and with both versions recoverable. This is different from typical of speech, which is a question for the Switchboard comparison. Either answer may be “unsure”, and I compute the rates over yes and no answers only. No sampled item was used to develop the prompt.

5.3 Switchboard comparison

Under construction.

I compare the synthesised items with the RSR in NXT Switchboard. Each measure looks at the head word of the reparandum and of the repair, and follows SPACER, which describes single-word substitution repairs from Switchboard in the same way.

- Word frequency: how common each word is in English, as its Zipf frequency in SUBTLEXus.
- Semantic distance: how far apart the two words are in meaning, as the cosine distance between their GloVe vectors.
- Phonetic distance: how different the two words sound, as the edit distance between their pronunciations in the CMU dictionary.
- Surprisal: how unexpected each word is in its place, as scored by a language model given the same left context, which ends before the repair site.

The comparison is descriptive. This project is about a meaning representation for RSR and a parser for it, not about how closely the data matches natural speech. So, I will not remove data for being unlike Switchboard.

6 Results

6.1 Quality by category

Table 4 sets each category against the thresholds.

Table 4: Automatic checks and manual annotation by category. Check rates are over answered items, from one run per category (5,984 items for the first six, 377 for edge label). Annotation counts are yes / (yes + no) on 15 items per category (34 for edge label), with unsure answers left out. Edge label was annotated on a run with the same prompt as its check rate.

category	passed checks	correct in meaning	natural
content word	93.2%	14/14	14/14
name	95.5%	11/11	14/14
number	94.6%	13/13	10/11
temporal	96.9%	15/15	14/14
tense	96.1%	15/15	14/14
forwarding	96.3%	4/4	11/12
edge label	97.1%	32/33	**14/29**
threshold	*95%*	*90%*	*80%*

All six categories other than edge label pass both annotation thresholds. The samples are small, though. With 15 items even a perfect score is compatible with a true rate as low as 80% (95% Wilson interval), so the annotation shows that no category is clearly failing but cannot rank them.

Content word and number fall just short of the check threshold. For content word this is due to synonyms, which only this category can produce; without them it passes 96.5%. Since failing items are dropped, the shortfall costs items but does not lower the quality of those kept. The model declined fewer than ten of nearly 6,000 sentences. What the failures consist of is more telling.

> Ill-formed SBN is rare (30 items). More often the model changed something the repair did not ask for, for instance by putting the two versions the wrong way round (“covered in pigeon, crow poop” for a gold sentence about pigeon poop). The model finds it harder

> to change exactly what the repair requires than to write well-formed SBN.

6.2 Edge label: the exception

Edge label was the hardest category to generate. It is the only one that does not meet the thresholds through generation and automatic checks alone. Its repairs pass the checks (97.1%) and are almost always correct in meaning, but many do not sound natural. Of the 988 edge-label items that passed the checks, I judged only 495 natural, about half.

No simple rule separates the natural items from the unnatural ones. Verbs that express a direction rarely allow a natural swap (e.g., "A lion escaped to, from the zoo"). For other verbs the judgement depends on tense and aspect. "Tom has gone from, to Boston" sounds also odd because "has gone" is close to a fixed expression for having left. No list of verbs could capture this.

Several rounds of prompt revision did not raise this share. The only way to a usable set was therefore to read every item that passed the checks and keep the natural ones. I expect that I need to do the same for dev/text sets.

7 Limitations and next steps

The evidence behind this answer has at least three main limits.

- **One annotator.** All annotation and the edge-label selection rest on my intuition. Considering that 1. I am not a native speaker of English, 2. intra-annotator drift exists, and 3. I only annotated the dataset once, the quality of the dataset can be questionable.
- **Samples and upper bounds.** Apart from edge label, no category was read in full. There might exist data that in fact wrong but can pass the structural check.
- **Written source.** PMB sentences are short written text, and the repairs are constructed, not observed. Natural here means acceptable to a fluent speaker, which the Switchboard comparison can complement but not replace.

Three steps come next. I will first add editing expressions ("I mean", "no", "no wait", "actually") by script to a fixed share of the items (10%), which gives matched pairs of the same repair with and without an expression. "Sorry" is kept back for the test set as an unseen expression.

After that, I plan to mix the synthesised RSR data with PMB Train Gold and upload it to HuggingFace as the train set.

I will then build the dev and test sets from PMB's own splits with the same prompts and checks, after completing the Switchboard comparison.

7.1 Questions for my supervisors

1. Should dev and test be annotated more heavily than train, for example in full? They are smaller, and an error there affects the evaluation directly.

8 Appendix

8.1 Examples per category

Table 5: Examples from the training set, drawn at random.

category	examples
content word	My wife is a good baker, cook. / They hugged, kissed inside the planetarium.
name	Were these pictures taken in France, in Germany? / I do a training at General Motors, Ford.
number	My watch is three, five minutes fast. / Tom was 25, 30 years old.
temporal	I'll pick you up at 3:30, 2:30. / We have lunch at eleven, midday.
tense	Tom loved, loves tulips. / He bows, bowed to the Queen.
forwarding	Jim passed out, Tom passed out from the heat. / John slapped, I slapped his face.
edge label	There is a glass under, on the table. / She used to live near, in Tel Aviv.

8.2 Failures per category

Table 6: Failures by category and check group for the first six categories (an item can fail more than one group).

category	answered	well-formed graph	graph returns to gold	sentence and graph agree	site within the category	a repair, not a synonym
content word	3,341	15	78	78	19	113
name	759	3	10	15	11	0

category	answered	well-formed graph	graph returns to gold	sentence and graph agree	site within the category	a repair, not a synonym
number	597	4	21	12	5	0
temporal	228	1	4	3	1	0
tense	760	5	8	22	4	0
forwarding	299	2	6	6	5	0